PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 107	Y. B. Chen	Discrete Mathematics and Statistics	6499

2025/26 SEMESTER 1 – Assessment II

BACHELOR DEGREE – Year 2

DISCRETE MATHEMATICS AND STATISTICS

DEADLINE: 17 December 2025 at 11:59 pm

INSTRUCTIONS TO CANDIDATES

1. **The Assessment should be done individually.**
2. **Total marks available are 100, accounting for 10% of the overall module marks.**
3. **The number in the column on the right indicates the marks for each question.**
4. **Answer all questions.**
5. **Answers should be written in English.**
6. **Relevant and clear steps should be included in your answers**
7. **Your solutions should be submitted electronically through the Learning Mall via the submission link.**
8. **The naming of Report (in pdf) is as follows: CPT107_StudentID_003.pdf (e.g. , CPT107_12345678_003.pdf)**
9. **Answers can also be handwritten, fully and clearly scanned or photographed for submission as one single PDF document through the Learning Mall via the submission link.**

Notes:

- **To obtain full marks for each question, relevant and clear steps must be included in the answers.**
- **Partial marks may be awarded depending on the degree of completeness and clarity.**

QUESTION I: Functions and PHP (18 marks)

1). Let $f: \mathbb{Q} \to \mathbb{Q}$, check whether $f(x) = \frac{x+5}{3x-7}$ and $f^{-1}(x) = \frac{7x+5}{3x-1}$ are inverses.

(2 marks)

2). Let $S = \{x | x \in \mathbb{R} \ and \ x \geq 1\}$ and $T = \{x | x \in \mathbb{R} \ and \ 0 < x \leq 1\}$.

Find a function $f: S \to T$ that is a bijection. (4 marks)

3). Show that in a party there are always two persons who have shaken hands with the same number of persons. (2 marks)

4). Let $f: \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 3x^2 + 2$ and $g: \mathbb{R} \to \mathbb{R}$ be given by $g(x) = 4x - 5$.
Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$ and $(g \circ g)(x)$.

(4 marks)

5). Find the composite function g ∘ f , where:

$$f: \mathbb{R} \to \mathbb{R}, f(x) = \begin{cases} x-2, if\ x\ \geq 1 \\ x^3, if\ x < 1 \end{cases}$$

$$g: \mathbb{R} \to \mathbb{R}, g(x) = \begin{cases} \dfrac{x+4}{3}, if\ x \geq 0 \\ |x+1|, if\ x < 0 \end{cases}$$

(6 marks)

QUESTION II: Logic (31 marks)

1). Consider the signature *S = {Cousin, Male, Female, jessie, carol, paul}* consisting of a binary predicate symbol *Cousin*, two unary predicate symbols *Male* and *Female*, and three constant symbols *jessie*, *carol* and *paul*. Assume that these symbols have the following meaning:

Cousin means "*is a cousin of*" (i.e., *Cousin (a, b)* states *a* is a cousin of *b*).

Male means "*is male*" (i.e., *Male(a)* states *a* is male).

Female means "*is female*" (i.e., *Female(a)* states *a* is female).

jessie, carol and *paul* refer to "Jessie", "Carol" and "Paul", respectively.

Translate the following sentences into S-formulae; that is, for each of the following sentences provide an S-formula that expresses the sentences:

(a) Jessie is a cousin of Carol. (3 marks)

(b) Paul has a female cousin. (3 marks)

(c) All cousins of Paul are also cousins of Carol. (3 marks)

(d) Jessie has at least 2 male cousins. (3 marks)

2). If statement *q* has the truth value 1 (TRUE), determine all truth value assignments for the statements *p, r*, and *s* for which the true value of the statement

$$(q \to ((\neg p \vee r) \wedge \neg s)) \wedge (\neg s \to (\neg r \wedge q))$$

is 1 (TRUE). (5 marks)

3). Determine the truth value of each of the following statements where U = {1, 2, 3} is the domain of discourse (a set of integer numbers):

(a) $\exists x\ \forall y\ (x^2 < y+1)$; (2 marks)

(b) $\forall x\ \forall y\ (x^2+y^2 < 12)$. (2 marks)

Explain the answers.

4). Without using any truth tables, prove or disprove that $\neg(p \wedge q) \vee (\neg p \wedge q) \equiv \neg p$.

(2 marks)

5). Let p denote "*He is rich*" and let q denote "*He is happy*." Write each statement in symbolic form using p and q. Note that "*He is poor*" and "*He is unhappy*" are equivalent to ¬p and ¬q, respectively.

(a) If he is rich, then he is unhappy. (2 marks)

(b) He is neither rich nor happy. (2 marks)

(c) It is necessary to be poor in order to be happy. (2 marks)

(d) To be poor is to be unhappy. (2 marks)

QUESTION III: Combinatorics (23 marks)

(Please use P(n, r) rather than A_n^r to denote Permutations and C(n, r) rather than C_n^r to denote Combinations.)

1). A student council consists of 15 students.

a). In how many ways can a committee of **six** be selected from the membership of the council?

(2 marks)

b). There are **two** council members from the same major. They are not permitted to serve together on a committee. How many ways can a committee of **six** be selected from the membership of the council? (2 marks)

c). **Two** council members always insist on serving on committees together. If they can not serve together, they will not serve at all. How many ways can a committee of **six** be selected from the council membership? (2 marks)

d). Suppose the council contains **eight** men and **seven** women.

(i) How many committees of **six** contain **three** men and **three** women? (2 marks)

(ii) How many committees of **six** contain at least **one** woman? (2 marks)

e). Suppose the council consists of ***three*** *freshmen*, ***four*** *sophomores,* ***three*** *juniors*, and ***five*** *seniors*. How many committees of **eight** contain **two** representatives from each class?

(2 marks)

2). If one quarter of all subsets of size 5 of the set {1, 2, 3, …, m} contains the element 3, determine *m*. (4 marks)

3). How many 5-digit numbers can be formed from the integers 1, 2, ..., 9 if no digit can appear more than twice? (For instance, 41434 is not allowed.) (4 marks)

4). Let A = {1, 2, ..., n + 1} and B = {1, 2, ..., n}. Find the number of functions f: A→B that are surjective. (3 marks)

QUESTION IV: Probability (28 marks)

1). Ten people (five men and five women) are standing in line. Assume that all possible ways in which they might line up are equally likely.

(a) What is the probability that they appear in line in alphabetical order by name? Please assume no two of them have the same English name. (3 marks)

(b) What is the probability that all the women precede all the men? (3 marks)

(c) What is the probability that between any two women there are no men? (3 marks)

(d) What is the probability that they alternate by gender in the line? (3 marks)

2). 50 butterflies are captured from a previously unexplored rainforest. 20 of these butterflies have stripes, 31 have spots and 14 have dots. 16 have stripes and spots, 9 have spots and dots and 5 have all three features. 12 of the 50 butterflies have none of these three features.

(a) What is the probability that a butterfly randomly selected from the sample has stripes and dots? (3 marks)

(b) What is the probability that a butterfly randomly selected from the sample has at least two of these three features? (3 marks)

3). When a pair of balanced dice is rolled and the sum of the numbers showing face up is computed, the result can be any number from 2 to 12, inclusive.
What is the expected value of the sum?

(4 marks)

4). A man fires at a target n = 6 times and hits it k = 2 times,

(a) List the different ways that this can happen, (3 marks)

(b) How many ways are there? (3 marks)

END OF EXAM PAPER